Analytic relationship of relative synchronizability to network structure and motifs

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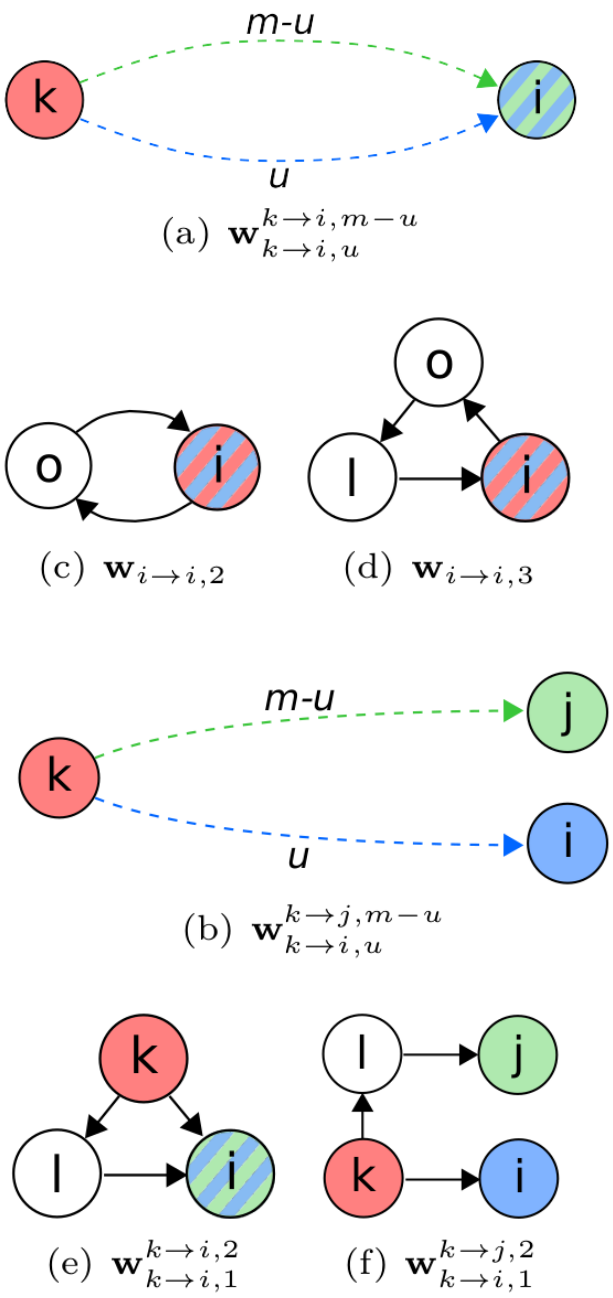
Faculty of Engineering



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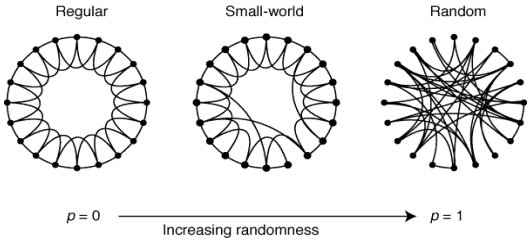
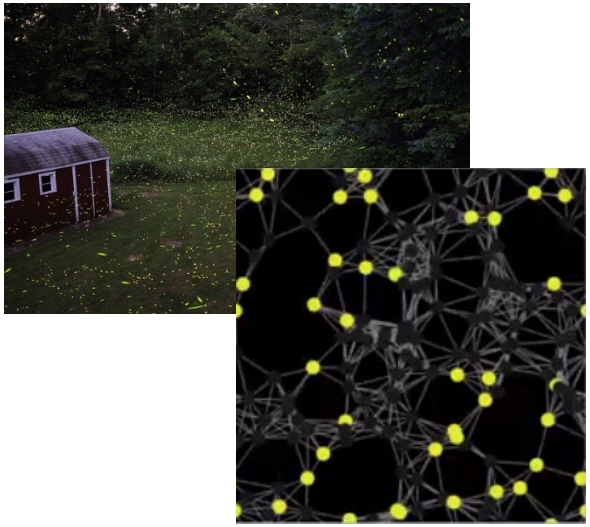
Relating network structure to dynamics

- Dynamics remains "*much less well understood*" than characterising structure alone.
  - The problem is simply more diverse.
- Ideally we need a common framework to understand the dynamic capabilities of various network structures.
- Specifically, we want to understand:
  - the general dynamic properties of structure;
  - how different dynamic processes relate to eachother.

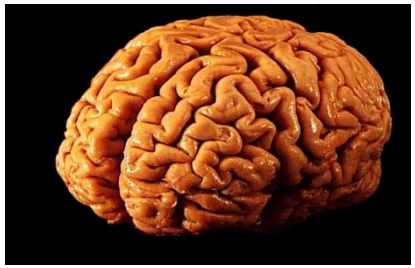


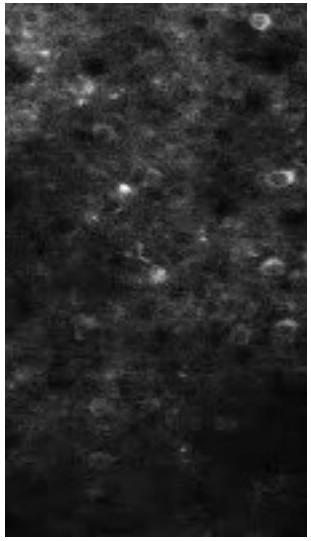


D. J. Watts and S. H. Strogatz, Nature 393, 440 (1998)
M. Mitchell, Artificial Intelligence 170, 1194 (2006).
A.-L. Barabsi, Science 325, 412 (2009).

Synchronization

- A self-organising phenomenon in complex systems, where interacting oscillators mutually entrain their states.
- Occurs in many natural and man-made systems





Strogatz S (2003) Sync: The Emerging Science of Spontaneous Order. (Hyperion Books, New York)
https://netlogoweb.org/launch#https://netlogoweb.org/assets/modelslib/Sample%20Models/Biology/Fireflies.nlogo
https://www.youtube.com/watch?v=ZGvtnE1Wy6U&pp=ygUOZmlyZWZsaWVzIHN5bmM%3D
https://www.youtube.com/shorts/4GleKfxW288

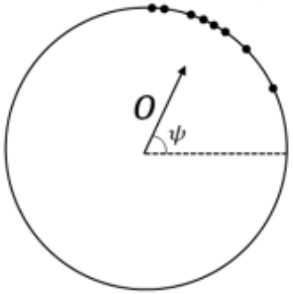
Synchronization

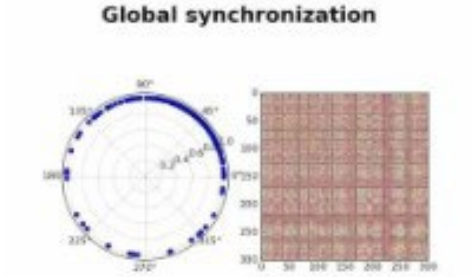
- **Kuramoto model** made sync popular in complex systems
  - Population of P oscillators, ω_X natural frequency and ϑ_X phase, coupled by strength K in (perhaps) a network A_{XY}, with governing equation:

$$\dot{\theta}_X(t) = \omega_X + K \sum_{Y=1}^{P} A_{XY} \sin\left(\theta_Y(t) - \theta_X(t)\right)$$

  - Order o defined as centroid of phases with ψ average phase

$$o e^{i\psi} = \frac{1}{P} \sum_{X=1}^{P} e^{i\theta_X}$$



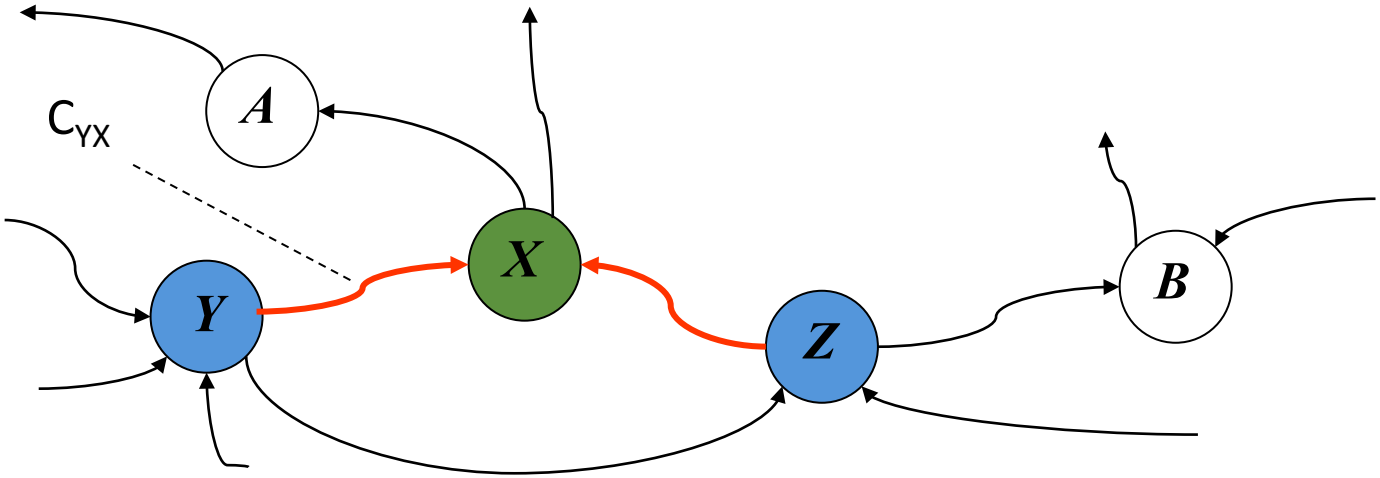


  - Early studies focussed on full coupling or mean field, and strength of coupling
  - Complex network structure will have non-trivial effects

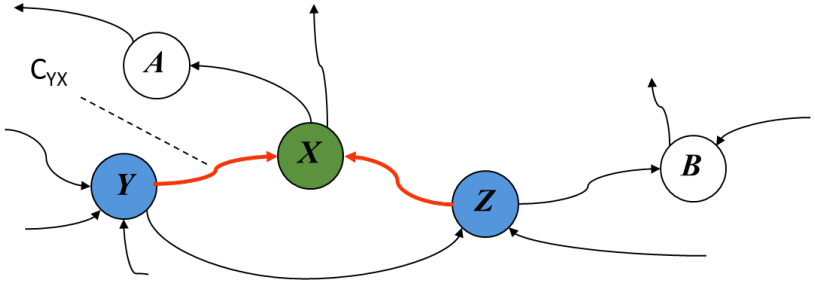
Kuramoto Y (1975) "Self-entrainment of a population of coupled non-linear oscillators", in International Symposium on Mathematical Problems in Theoretical Physics. (Springer Berlin / Heidelberg) Vol. 39, pp. 420–422.
Kuramoto Y (1984) Chemical Oscillations, Waves, and Turbulence. (Springer-Verlag).
https://www.youtube.com/watch?v=Vd-ww9PcZvI

Analytic relationship of relative synchronizability to network structure and motifs

- **Big picture aim**: Given a network structure $C=[C_{YX}]$, *analytically*:
  - **Determine whether** the network is synchronizable, and
  - **Rank the relative quality** of that synchronization, and
  - **Understand** how this relates to network structure,



Dynamics and sync conditions: scope



- Determine analytic results from the linear regime (weakly-coupled non-linear regime)
- For a weighted, directed network structure $C=[C_{YX}]$ (normalised Laplacian $L = I-C$) between N nodes:
  - Usually consider C with **synchronised state vector** ψ_0 = [1,1,1,…,1] as an eigenvector with $\lambda_0 = 1$

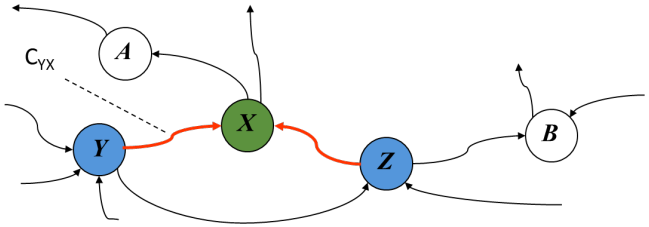
	Discrete (VAR)	Continuous (Ornstein-Uhlenbeck)
Dynamics	$\vec{x}(t+1) = \vec{x}(t)C + \vec{r}(t)$	$d\vec{x}(t) = -\vec{x}(t)(I-C)dt + d\vec{w}(t)$
Sync condition	$\lvert\lambda\rvert < 1$	$\mathrm{Re}(\lambda) < 1$ Except ψ_0

- Sync condition: $\sigma^2(\infty) \to 0$ without ongoing noise

$$\sigma^2(t) = \sum_{i=1}^{N} (x_i(t) - \bar{x}(t))^2$$

- Interpretation: system is stable to disturbances orthogonal to ψ_0

Background



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Limited: heuristic, empirical or symmetric

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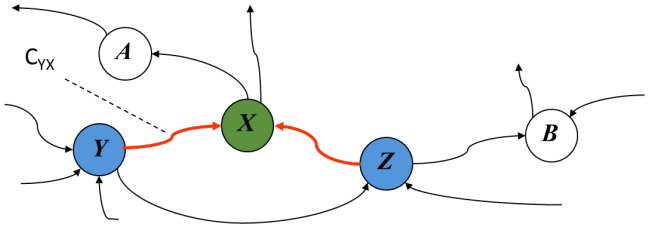
Sync condition	$\lvert\lambda\rvert < 1$	$\mathrm{Re}(\lambda) < 1$	Except ψ_0
Synchronizability increases with:	$\lambda_{min}/\lambda_{max}$ of L	λ_{max} of C	**Heuristic**

Atay FM, Jost J, Wende A (2004) Delays, connection topology, and synchronization of coupled chaotic maps. Physical Review Letters 92(14):144101.
Atay FM, Biyikoğlu T (2005) Graph operations and synchronization of complex networks. Physical Review E 72(1):016217
Li, X., Chen, G. "*Synchronization and Desynchronization of Complex Dynamical Networks: An Engineering Viewpoint*" , IEEE Transactions On Circuits and Systems I: Fundamental Theory and Applications, 50, no. 11 (2003): 1381–90.
Jost J, Joy MP (2001) Spectral properties and synchronization in coupled map lattices. Physical Review E 65(1):016201

Background



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Sync condition	$\lvert\lambda\rvert < 1$	$\mathrm{Re}(\lambda) < 1$ Except ψ_0

- Sync condition $\equiv \langle\sigma^2(\infty)\rangle$ remains finite (expected steady-state deviation under driving noise)
- Synchronizability: **empirically** evaluate $\langle\sigma^2(\infty)\rangle$

Korniss G, Novotny MA, Guclu H, Toroczkai Z, Rikvold PA (2003) Suppressing Roughness of Virtual Times in Parallel Discrete-Event Simulations. Science 299(5607):677–679
Atay FM, Jost J, Wende A (2004) Delays, connection topology, and synchronization of coupled chaotic maps. Physical Review Letters 92(14):144101.
Korniss G (2007) Synchronization in weighted uncorrelated complex networks in a noisy environment: Optimization and connections with transport efficiency. Physical Review E 75(5):051121

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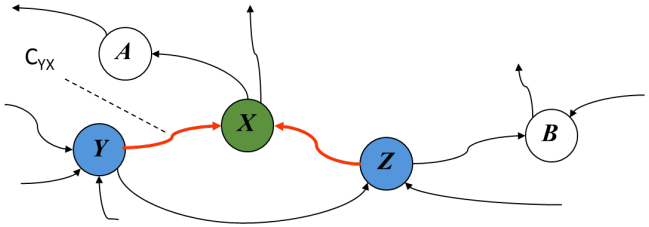

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Sync condition	$\lvert\lambda\rvert < 1$	$\mathrm{Re}(\lambda) < 1$ Except ψ_0

- Synchronizability: for **symmetric** networks:

$$\langle \sigma^2 \rangle = \frac{1}{2N} \sum_{v=1}^{N-1} \frac{1}{1-\lambda_v}$$

Mapped to our system

Korniss G (2007) Synchronization in weighted uncorrelated complex networks in a noisy environment: Optimization and connections with transport efficiency. Physical Review E 75(5):051121

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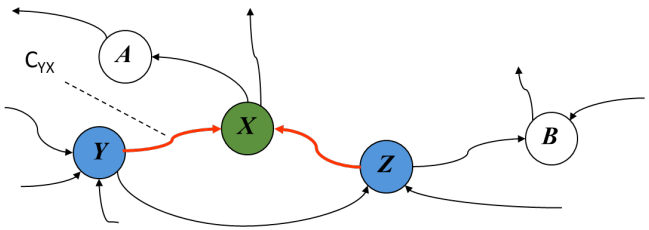

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  - Determine whether the network is synchronizable, and
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Limited: heuristic, empirical or symmetric

Special circumstances

	Discrete (VAR)	Continuous (Ornstein-Uhlenbeck)
Dynamics	$\vec{x}(t+1) = \vec{x}(t)C + \vec{r}(t)$	$d\vec{x}(t) = -\vec{x}(t)(I-C)dt + d\vec{w}(t)$
Sync condition	$\lvert\lambda\rvert < 1$	$\mathrm{Re}(\lambda) < 1$ Except ψ_0

Synchronizability and network structure

- Regular and small-world networks are less synchronizable (Atay et al., 2004);
- Full connectivity increases synchronizability, partionability decreases the closer the network becomes to bipartite (Atay and Karabacak, 2006)
- Relative synchronizability of various motifs in isolation (Moreno Vega et al., 2004)
- Synchronizability decreases with the spread of the eigenvalues (Korniss, 2007).
- Synchronizability decreases as network is easier to partition (Bauer et al., 2013)
- ...

Atay FM, Jost J, Wende A (2004) Delays, connection topology, and synchronization of coupled chaotic maps. Physical Review Letters 92(14):144101.
Atay FM, Karabacak O (2006) Stability of coupled map networks with delays. SIAM Journal on Applied Dynamical Systems 5(3):508–527.
Moreno Vega Y, Vázquez-Prada M, Pacheco (2004) Fitness for synchronization of network motifs. Physica A 343:279–287.
Korniss G (2007) Synchronization in weighted uncorrelated complex networks in a noisy environment: Optimization and connections with transport efficiency. Physical Review E 75(5):051121
Bauer F, Jost J (2013) Bipartite and neighborhood graphs and the spectrum of the normalized graph Laplacian. Communications in Analysis and Geometry 21(4):787–845

Analytic relationship of relative synchronizability to network structure and motifs

- Given a network structure $C=[C_{YX}]$, *analytically*:
  - Rank the relative quality of that synchronization, and
  - Understand how this relates to network structure, inc. motifs, and
  - Extend to dynamics with coupling delays.

	Discrete (VAR)	Continuous (Ornstein-Uhlenbeck)
Dynamics	$\vec{x}(t+1) = \vec{x}(t)C + \vec{r}(t)$	$d\vec{x}(t) = -\vec{x}(t)(I-C)dt + d\vec{w}(t)$

1. J.T. Lizier, F. Bauer, F. M. Atay, J. Jost, "Analytic relationship of relative synchronizability to network structure and motifs", *under submission.*
2. J.T. Lizier, F. M. Atay, J. Jost, "Relating synchronizability with delays to network structure and motifs", *in preparation.*

1. Write synchronisability in terms of covariance

- We express $<\sigma^2>$ in terms of network covariance matrix $\mathbf{\Omega}$:

$$\begin{aligned}\left\langle \sigma^2 \right\rangle &= \lim_{t\to\infty} \frac{1}{N} \sum_{i=1}^{N} \left\langle (x_i(t) - \bar{x}(t))^2 \right\rangle, \\ &= \lim_{t\to\infty} \frac{1}{N} \mathrm{trace}\left(\left\langle (U^T \vec{x}(t)^T)(\vec{x}(t)U) \right\rangle \right), \\ &= \frac{1}{N} \mathrm{trace}\left(U^T \Omega U \right), \\ &= \frac{1}{N} \mathrm{trace}\left(\Omega_U \right),\end{aligned}$$

- Where:

$$\Omega = \lim_{t\to\infty} \left\langle \vec{x}(t)^T \vec{x}(t) \right\rangle$$

Network covariance matrix

$$\vec{x}U = \vec{x} - \bar{x}\vec{\psi}_0,$$

U is the unaveraging operator

$$\Omega_U \triangleq \lim_{t\to\infty} \left\langle U^T \vec{x}(t)^T \vec{x}(t) U \right\rangle,$$

Network covariance in space orthogonal to sync

$$\Omega_U = U^T \Omega U$$

When is Ω well defined

- If we can write down $\mathbf{\Omega}_U$ then we can write down $<\sigma^2(\infty)>$

2. Express covariance in terms of network structure

- Barnett et al. express covariance as a power series:

	Discrete (VAR)	Continuous (Ornstein-Uhlenbeck)
Dynamics	$\vec{x}(t+1) = \vec{x}(t)C + \vec{r}(t)$	$d\vec{x}(t) = -\vec{x}(t)(I-C)dt + d\vec{w}(t),$
Solution for $\mathbf{\Omega}$	$\sum_{u=0}^{\infty} (C^u)^T C^u$	$\sum_{m=0}^{\infty} 2^{-m} \sum_{u=0}^{m} \binom{m}{u} (C^u)^T C^{m-u}$

- But this is only valid for $|\lambda| < 1$
  - i.e. *not* when ψ_0 is an eigenvector with $\lambda_0 = 1$
  - Cannot plug directly into $\Omega_U = U^T \Omega U$ when we have ψ_0
  - Need to derive for $\mathbf{\Omega}_U$ directly

Barnett L, Buckley CL, Bullock S. "Neural complexity and structural connectivity". Phys. Rev. E 79, 051914, 2009

2. Express covariance in terms of network structure

- We extend Barnett et al.'s solution in presence of ψ_0:

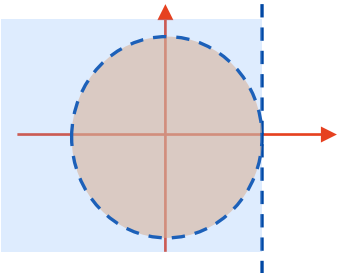
	Discrete (VAR)	Continuous (Ornstein-Uhlenbeck)
Dynamics	$\vec{x}(t+1) = \vec{x}(t)C + \vec{r}(t)$	$d\vec{x}(t) = -\vec{x}(t)(I-C)dt + d\vec{w}(t),$
Solution for $\mathbf{\Omega}_U$	$\sum_{u=0}^{\infty} U(C^u)^T C^u U$	$\sum_{m=0}^{\infty} 2^{-m-1} \sum_{u=0}^{m} \binom{m}{u} U(C^u)^T C^{m-u} U$

- Now is valid for $|\lambda| < 1$, except for $\lambda_0 = 1$ corresponding to ψ_0 when it is an eigenvector:
  - Matches form plugging directly into $\Omega_U = U^T \Omega U$ (so solution holds without zero mode)
  - Valid for all C which synchronise in discrete-time
  - Works for a subset in continuous-time

2. Solution for $<\sigma^2>$ in terms of network structure

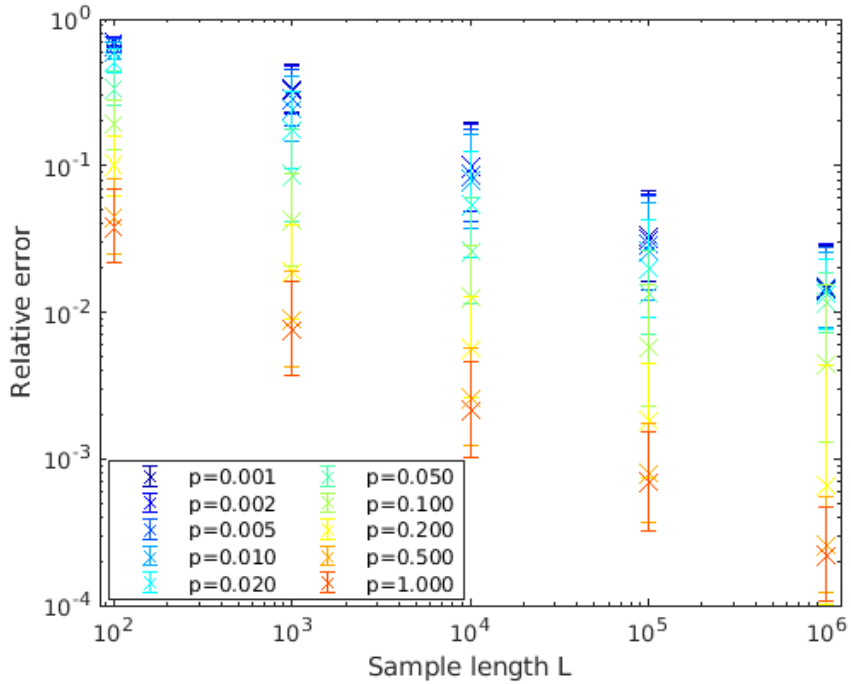
	Discrete (VAR)	Continuous (Ornstein-Uhlenbeck)
Dynamics	$\vec{x}(t+1) = \vec{x}(t)C + \vec{r}(t)$	$d\vec{x}(t) = -\vec{x}(t)(I-C)dt + d\vec{w}(t),$
Solution for $<\sigma^2>$	$\frac{1}{N}\sum_{u=0}^{\infty} \mathrm{trace}\left(U(C^u)^T C^u U\right)$	$\sum_{m=0}^{\infty} \frac{2^{-m-1}}{N} \sum_{u=0}^{m} \binom{m}{u} \mathrm{trace}\left(U(C^u)^T C^{m-u} U\right)$

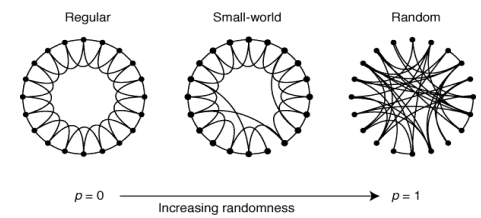
- We can now **rank the quality of synchronisation** fully analytically
  - not heuristic, empirical or symmetric or diagonalizable
  - is valid for $|\lambda| < 1$, except for $\lambda_0 = 1$ corresponding to ψ_0 when it is an eigenvector:
    - Valid for all C which synchronise in discrete-time
    - Valid for a subset of C which synchronise in continuous-time



Numerical validation

Error between $<\sigma^2>$ evaluated empirically and analytically





d=4, p randomises source of edges, C_{ii}=0.5, C_{ij}=0.5/d for $i \neq j$

- Empirical results converge exponentially with L to analytic
  - Validates analytic results
  - Holds for across network types
    - Smaller relative error for random networks

Code available at https://github.com/jlizier/linsync

Ranked quality, can we relate to structure?

- Given a network structure $C=[C_{YX}]$, *analytically*:
  - Rank the relative quality of that synchronization, and
  - **Understand how this relates to network structure, inc. motifs,** and
  - Extend to dynamics with coupling delays.

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Solution for $<\sigma^2>$	$\frac{1}{N}\sum_{u=0}^{\infty} \mathrm{trace}\left(U(C^u)^T C^u U\right)$	$\sum_{m=0}^{\infty} \frac{2^{-m-1}}{N} \sum_{u=0}^{m} \binom{m}{u} \mathrm{trace}\left(U(C^u)^T C^{m-u} U\right)$

- We'll utilise identity:

$$\mathrm{trace}(U^T A U) = \sum_i A_{ii} - \frac{1}{N}\sum_{i,j} A_{ij}$$

Derivation for continuous-time

Expanding $\mathrm{trace}(U^T A U)$

$$\langle \sigma^2 \rangle = \sum_{m=0}^{\infty} \frac{2^{-m-1}}{N} \sum_{u=0}^{m} \binom{m}{u} \times \left(\sum_i ((C^u)^T C^{m-u})_{ii} - \frac{1}{N} \sum_{i,j} ((C^u)^T C^{m-u})_{ij} \right),$$

Expanding matrix product

$$\langle \sigma^2 \rangle = \sum_{m=0}^{\infty} \frac{2^{-m-1}}{N} \sum_{u=0}^{m} \binom{m}{u} \times \left(\sum_{i,k} C^u_{ki} C^{m-u}_{ki} - \frac{1}{N} \sum_{i,j,k} C^u_{ki} C^{m-u}_{kj} \right),$$

$$\langle \sigma^2 \rangle = \sum_{m=0}^{\infty} \frac{2^{-m-1}}{N} \sum_{u=0}^{m} \binom{m}{u} \times \sum_{i,k} \left(\mathbf{w}^{k \to i, m-u}_{k \to i, u} - \frac{1}{N} \sum_j \mathbf{w}^{k \to j, m-u}_{k \to i, u} \right),$$

Weighted dual-walk counts

$$\mathbf{w}^{a \to e, M_2}_{a \to b, M_1} = \mathbf{w}_{a \to b, M_1} \mathbf{w}_{a \to e, M_2}, \quad = (C^{M_1})_{ab} (C^{M_2})_{ae},$$

Deviation from sync for continuous time

$$\langle \sigma^2 \rangle = \sum_{m=0}^{\infty} \frac{2^{-m-1}}{N} \sum_{u=0}^{m} \binom{m}{u} \times \sum_{i,k} \left(\mathbf{w}_{k\to i,u}^{k\to i,m-u} - \frac{1}{N} \sum_{j} \mathbf{w}_{k\to i,u}^{k\to j,m-u} \right)$$

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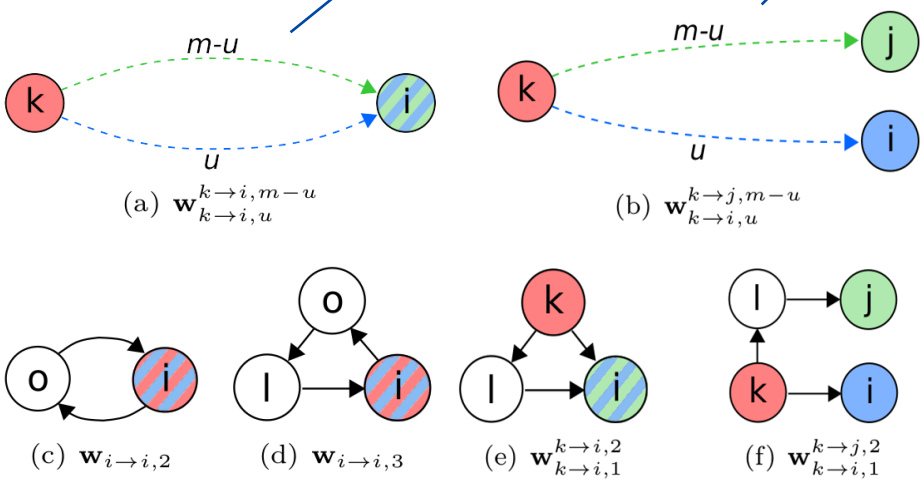


- $\langle\sigma^2\rangle$ asks if weighted walks of length *m-u* from *k*, which has a (non-zero weighted) walk count *k*→*i* of length *u*, preferentially end at node *i*.
- Bad for sync to have weighted count of closed walks as a high proportion of such dual walks

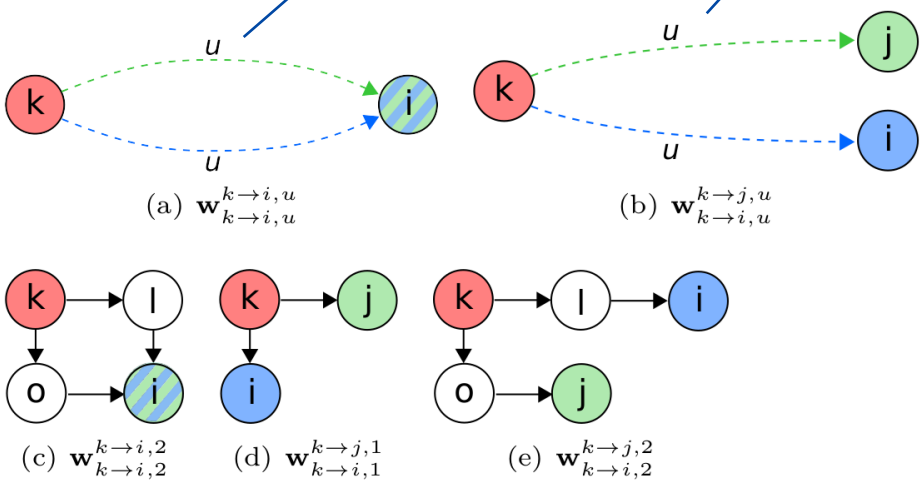
J.T. Lizier, F. Bauer, F. M. Atay, J. Jost, "Analytic relationship of synchronization ability to network structure and motifs", *under submission*

Deviation from sync for discrete time

$$\langle \sigma^2 \rangle = \frac{1}{N} \sum_{u=0}^{\infty} \left(\sum_{i,k} \mathbf{w}_{k\to i,u}^{k\to i,u} - \frac{1}{N} \sum_{i,j,k} \mathbf{w}_{k\to i,u}^{k\to j,u} \right)$$

Weighted dual-walk counts

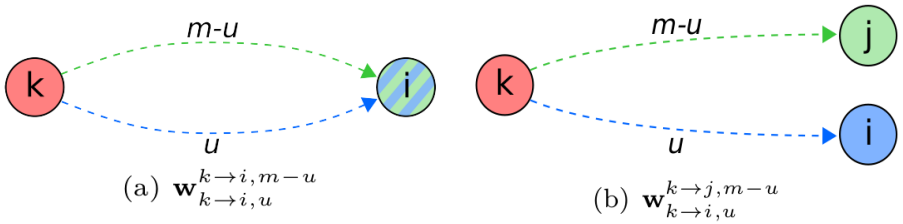
$$\mathbf{w}_{a\to b,M_1}^{a\to e,M_2} = \mathbf{w}_{a\to b,M_1} \mathbf{w}_{a\to e,M_2}, \\ = (C^{M_1})_{ab} (C^{M_2})_{ae},$$



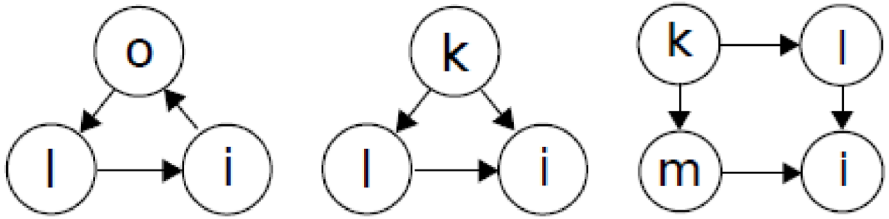
- $\langle\sigma^2\rangle$ asks if weighted walks of length *u* from *k*, which has a (non-zero weighted) walk count *k*→*i* of length *u*, preferentially end at node *i*.
- Bad for sync to have weighted count of closed walks as a high proportion of such dual walks (of *same* length)

J.T. Lizier, F. Bauer, F. M. Atay, J. Jost, "Analytic relationship of synchronization ability to network structure and motifs", *under submission*

Sync discussion



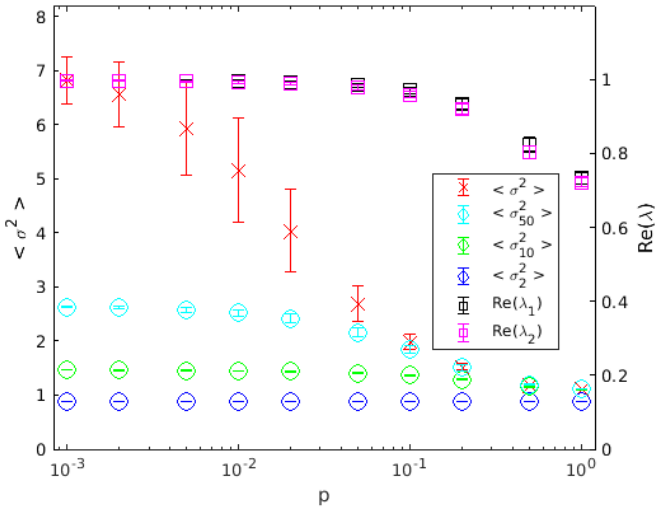
- Why are high proportions of closed walk motifs detrimental to sync?
  - They introduce reinforcement of noise
    - Operates differently for discrete vs continuous time
  - They store traces of dynamics in one part of the network only
  - In contrast to other dual-walks, they hinder dissemination or transfer of perturbations.
- These are information storage motifs
- Sync dynamics as another reason for these motifs' prevalence

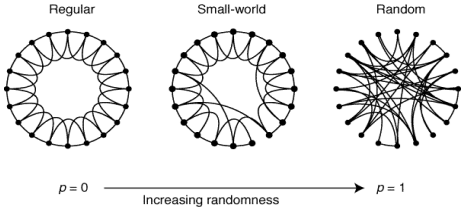


J. T. Lizier, F. M. Atay, and J. Jost, Information storage, loop motifs, and clustered structure in complex networks, Physical Review E 86, 026110 (2012)

Sync: insights

- Explains why **small-world and regular** networks are less synchronisable
- Same for **symmetric** networks
- Stronger contribution of shorter motifs - but all required
- Caveats: about proportions, and be careful with negative weights
- Better than heuristics based on eigenspectrum





d=4, p randomises source of edges, C_{ii}=0.5, C_{ij}=0.5/d for $i \neq j$

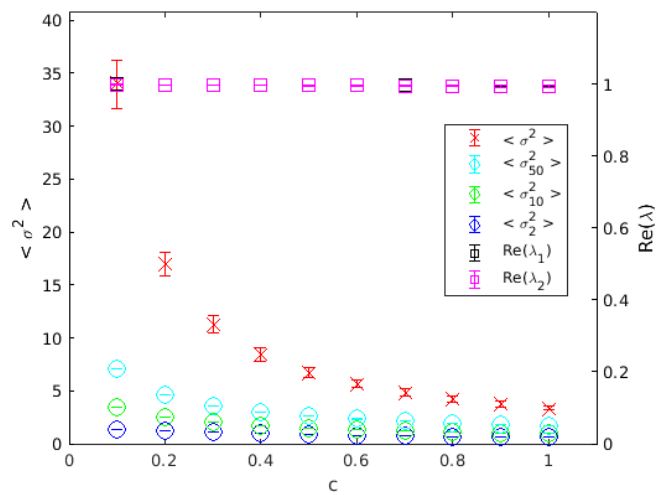
Code available at https://github.com/jlizier/linsync

Sync: insights

- Better than heuristics based on eigenspectrum
- Can also interpret in terms of difference between auto- and cross-covariance

$$\langle \sigma^2 \rangle = \frac{1}{N} \sum_i \Omega_{ii} - \frac{1}{N^2} \sum_{i,j} \Omega_{ij}$$

- Modularity will enhance sync inside module but decrease it across network



d=4, *p*=0.001,
C_{ii}=1-c, C_{ij}=c/*d* for *i*≠*j*

- Given a network structure $C=[C_{YX}]$, *analytically*:
  - Rank the relative quality of that synchronization, and
  - **Understand how this relates to network structure, inc. motifs,** and
  - Extend to dynamics with coupling delays.

Code available at https://github.com/jlizier/linsync

A. C. Schwarze and M. A. Porter, Motifs for processes on networks, SIAM Journal on Applied Dynamical Systems 20, 2516 (2021)

Summary

- Relating network structure to dynamics has remained poorly understood
  - Synchronisation has been a canonical problem here
- Given a network structure $C=[C_{YX}]$, *analytically*:
  - Rank the relative quality of that synchronization, and
  - Understand how this relates to network structure, inc. motifs, and
  - Extend to dynamics with coupling delays.
- Much scope for expanding these results!

Discrete time

Questions

THE UNIVERSITY OF SYDNEY